Long-Term Impacts of Early-Life Cash Transfers

A Causal Forest Approach

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Replication Archive

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Executive Summary

Objective: Re-evaluate the historical U.S. Mothers' Pension program (1911–1935) to uncover long-term treatment effect heterogeneity on child longevity.

Methodology:

- Applied Generalized Random Forests (GRF) to administrative and census data.
- Utilized a doubly-robust, orthogonalized machine learning framework.
- Addressed confounding via propensity score trimming and county-level clustering.

Main Finding:

- Program acceptance increased recipient lifespan by an average of **1.55 years**.

Treatment Heterogeneity:

- Omnibus and covariate-specific tests revealed significant spatial variation.
- Benefits were strongest in areas with lower baseline public spending on education and charities, highlighting the marginal value of targeted transfers.

Introduction

Empirical Research Question

- What are the long-term effects of early-life cash transfers on longevity?
- How does treatment heterogeneity manifest across individuals and regions?

Motivation and Relevance

- Cash transfers play a central role in modern social policy.
- Ongoing debates about their effectiveness and optimal design.
- Average Treatment Effects may mask important heterogeneity.
- Understanding variation helps design more effective policies.

Overview of Approach

- Use historical data from the U.S. Mothers' Pension (MP) program. It provides a valuable natural experiment for studying long-term impacts.
- Apply Causal Forests, a state-of-the-art machine learning method.
- Estimate both Average Treatment Effects (ATEs) and Conditional Average Treatment Effects (CATEs).
- Focus on how impacts vary across individual characteristics.

Early-Life Poverty and Cash Transfers

- Over 20% of U.S. children lived in poverty as recently as 2010.
- Early-life poverty has strong negative effects on education, health, and mortality (Almond and Currie, 2011).
- Parental income is a robust predictor of children's educational attainment and adult health outcomes (Barrow and Schanzenbach, 2012; Reardon, 2011; Case et al., 2002).
- However, evidence on whether direct cash transfers improve long-term outcomes remains mixed (Currie, 1998).

Challenges and Limitations

Why Cash Transfers May Have Limited Impact

- Transfer amounts may be too small to substantially improve well-being.
- Parents may not allocate funds optimally due to lack of information or other constraints (Dizon-Ross, 2014).
- Behavioral responses: changes in labor supply, fertility, or marriage decisions (Moffitt, 1998).

Methodological Challenges

- Difficulties constructing credible counterfactuals.
- Scarcity of long-term data linking recipients and control groups (Aizer et al., 2016).
- Overlapping programs (e.g., Medicaid, food stamps) complicate causal identification.

The Original Study: Aizer et al. (2016)

Overview

- Studied the U.S. Mothers' Pension program (1911–1935), first public cash assistance.
- Tracked 16,000 boys, linking applications to Census, WWII, and death records.
- Compared accepted vs. rejected applicants with similar baseline conditions.

Key Findings

- +1 year lifespan on average.
- Higher education (+0.34 years), better health, higher incomes.
- Strongest gains among the poorest families.

Limitations

- No data on girls.
- Excluded African-Americans.
- Relied mainly on ATEs and simple subgroup comparisons.

Causal Machine Learning

Why Causal Forests?

- Traditional methods mainly estimate ATEs.
- Causal forests directly estimate CATEs across individuals and subgroups (Athey and Wager, 2018; Athey et al., 2019).
- Especially relevant for detecting heterogeneity in long-term impacts of cash transfers.

Advantages

- Correct for confounding via propensity scores and orthogonalization.
- Address clustered errors (e.g., counties) with cluster-robust estimation.
- Sample-splitting ("honest estimation") prevents overfitting (Athey and Wager, 2019).

Implications

- Move beyond average effects.
- Reveal how impacts vary by family background, region, or maternal status.
- Inform more targeted and effective policies.

Identification Strategy

- Leverage quasi-experimental variation in the MP program.
- Compare children of accepted vs. rejected applicants.
- All applicants were initially deemed eligible; rejection often due to minor differences or local practices (Aizer et al., 2016).
- Under conditional unconfoundedness, treatment assignment approximates random variation:

$$\{Y_i(1), Y_i(0)\} \perp MP_i \mid X_i$$

where:

- $Y_i(1), Y_i(0)$ — log age at death.
- MP_i — treatment indicator (accepted $= 1$).
- X_i — rich vector of pre-treatment covariates.

Strengths and Limitations

Strengths

- Rich baseline data on family and economic characteristics.
- Administrative process created quasi-random variation.
- Limited alternative assistance reduces confounding.

Potential Threats

- Selection on unobservables (e.g., motivation, networks).
- Differential attrition affecting record linkage.
- Measurement error in treatment status.

Empirical Specification

Empirical Model

$$\log(\text{Age at Death}_i) = \alpha + \beta \cdot MP_i + \gamma' X_i + \delta_s + \lambda_c + \eta_t + \varepsilon_i$$

- δ_s, λ_c, η_t: state, county, and cohort fixed effects.
- β: the causal effect of MP cash transfers on log longevity.
- Given conservative comparison (rejected families often slightly better off), the estimates likely represent a **lower bound** on the true effect.

Robustness Strategies

1. Rich controls and fixed effects.
2. Cluster-robust standard errors (county level).
3. Orthogonalization via regression forests.
4. Sensitivity analyses (Rosenbaum, 2002).

Causal Forest: Goal

Goal

Estimate the Conditional Average Treatment Effect (CATE):

$$\tau(x) = \mathbb{E}\left[Y_i(1) - Y_i(0) \,\middle|\, X_i = x\right]$$

where:

- Y_i: log age at death
- W_i: indicator of MP acceptance
- X_i: covariates (child, family, cohort, region)

Estimation Steps

1. Orthogonalization

$$\hat{m}(X_i) = \mathbb{E}[Y_i \mid X_i = x], \quad \hat{e}(X_i) = \mathbb{E}[W_i \mid X_i = x]$$

$$\tilde{Y}_i = Y_i - \hat{m}(X_i), \quad \tilde{W}_i = W_i - \hat{e}(X_i)$$

Intuition: Remove variation explained by covariates to isolate the residual treatment effect.

2. R-Learner Objective

$$\hat{\tau}(\cdot) = \arg\min_{\tau} \left\{ \sum_{i=1}^{n} \left[\left(Y_i - \hat{m}^{(-i)}(X_i)\right) - \tau(X_i)\left(W_i - \hat{e}^{(-i)}(X_i)\right) \right]^2 + \Lambda_n\big(\tau(\cdot)\big) \right\}.$$

Intuition: Find the function that best predicts residualized outcomes from residualized treatments, balancing fit and complexity.

Estimation Details

3. Honest Estimation

- Sample splitting: one part used to grow trees (find splits), the other to estimate treatment effects.

Intuition: Prevents overfitting by separating model selection and estimation.

4. Tree Growing

- Recursive partitioning of covariate space to capture where treatment effects differ most.

Intuition: Trees adaptively detect heterogeneity across subgroups.

5. Inference

- CATE standard errors from variability across trees.
- ATE confidence intervals via influence functions.

Intuition: Use resampling and influence functions to quantify uncertainty.

Data Source and Sample

Historical dataset: Mothers' Pension program (1911–1935)

Coverage:

- 11 states (e.g., IL, OH, WI)
- Original records 80,000 children

Final Sample:

- Male children born 1900–1925
- Exact matches to Social Security Death Master File (1965–2012)
- N = 7,800 uniquely linked individuals

Rationale:

- Females excluded (name changes)
- Counties with both accepted/rejected applicants retained

Data Preparation and Variable Definition

First, the matched dataset was loaded and prepared.

Outcome Variable:

$$Y_i = \log(\text{age at death}_i)$$

Treatment Variable:

$$W_i = 1\{\text{MP application accepted}_i\}$$

Covariates (X_i) include:

- Mother: divorced, husband absent, marital status missing.
- Child: age, family name length, sibling dummies (2–8), min/max child age.
- County (1910 Census): SEI mean/std, urban share, widow share, single-mother share, poverty, female labor, child labor, farmland value.
- Matching: birth date missing.
- Fixed effects: county, state, cohort.
- State-year: wages, education laws, welfare/school/charity spending.
- Cohort variables: year of birth (linear/squared).

Model Specifications

Specifications Used for Robustness Checks:

Specification (Description)	**Definition**
Baseline (state and cohort fixed effects only)	X_i = State dummies + Cohort dummies
Extended (all controls, no county fixed effects)	X_i = Mother + Child + Matching + County characteristics + State-year characteristics + Cohort dummies
Full (with county fixed effects)	X_i = Mother + Child + Matching + County dummies + State-year characteristics + Cohort dummies
Alternative outcome (same covariates as Full model)	$Y_i = \log(\text{age at death}_{\text{SSA DOB}})$

Estimating the Propensity Score

Objective: Estimate the probability of MP program acceptance given observed covariates.

Method: Logistic regression: $\Pr(W_i = 1 \mid X_i)$

Only complete cases were used (no missing covariates).
Predicted propensity scores $\hat{e}(X_i)$ stored.

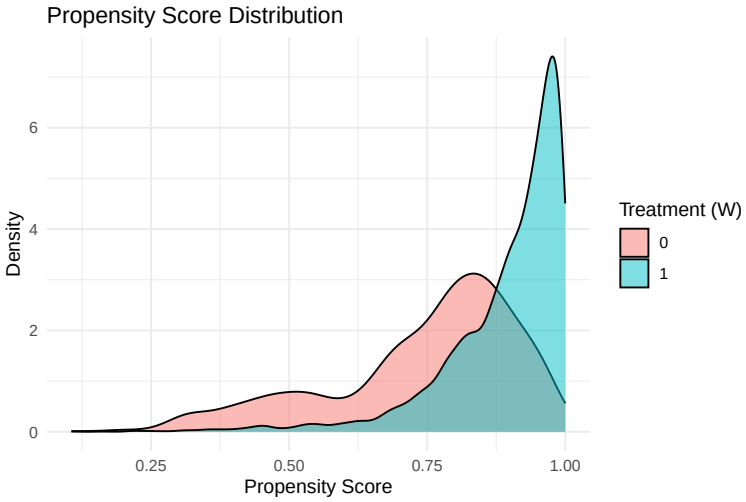
Summary Statistics: Min $= 0.11$, Median $= 0.91$, Mean $= 0.87$, Max $= 1.00$

Treatment Imbalance: Accepted: 6,877 (87.5%), Rejected: 983 (12.5%)

Interpretation

- Treated observations are concentrated near 1.0.
- Untreated are dispersed at lower scores.
- Limited overlap indicates the need for trimming.

Propensity Score Distribution



Propensity Score Distribution. Source: data from Aizer et al. (2016); author's own calculations.

Estimating the ATE Using Causal Forests

Estimated ATE using the causal forest method (`grf` package in R)

Results:

Model	ATE (95% CI)	Control ATE (95% CI)
Baseline	0.012 ($\pm$0.014)	0.016 ($\pm$0.014)
Extended	0.016 ($\pm$0.014)	0.015 ($\pm$0.014)
Full	0.015 ($\pm$0.013)	0.015 ($\pm$0.014)
Alt. Outcome	0.014 ($\pm$0.013)	0.014 ($\pm$0.014)

Implied lifespan gain (Full Model): **1.09 years** (95% CI: 0.1–2.09)

Interpretation and Notes

- All models estimated with `equalize.cluster.weights = FALSE`.
- Estimates are positive but smaller than in the original MP study ($\widehat{\text{ATE}} \approx 0.0182$).
- Several confidence intervals include zero.

Trimming Based on Propensity Score

Motivation: Limited common support observed in propensity score distribution.

Approach:

- Applied trimming to improve overlap between treated and control units.
- Trimming bounds set to 0.65 (lower) and 0.97 (upper) based on quantiles.

Results:

- 2,317 observations dropped.
- 5,542 observations retained for further analysis.

Interpretation

The trimmed sample shows substantially better overlap between treated and control groups, confirming appropriateness of trimming thresholds.

Propensity Score Distribution After Trimming

Propensity Score Distribution (Full vs Trimmed Sample). Source: data from Aizer et al. (2016); author's own calculations.

Causal Forest Estimation on Trimmed Data

Estimated the causal forest on the trimmed sample (propensity score trimming applied).

Main Specification: Full set of controls, county fixed effects.

Average Treatment Effect:

$$\widehat{\text{ATE}} = 0.021 \quad (95\% \text{ CI: } [0.005,\ 0.037])$$

Implied lifespan gain: **1.55 years** (95% CI: 0.36–2.79)

Selected Covariates:

Category	Covariates
Child characteristics	Age, sibling ages, number of siblings (3–8 children)
Maternal status	Divorced, never married, abandoned, institutionalized, missing marital status
State-level factors	Manufacturing earnings ratio, child labor laws, school entry age, continuation school requirements, assistance obligations
Public spending	Spending on social programs, charities, school spending, statutory allowances
Birth cohorts	Indicators for 1909, 1915, 1916

Omnibus Test for Treatment Effect Heterogeneity

To assess whether treatment effects vary across individuals, I conduct an *omnibus test* comparing ATEs above and below the median estimated CATE.

Approach:

- Split trimmed sample into high-effect and low-effect groups.
- Estimate ATE separately in each subgroup.
- Compare ATE difference.

Result:

$$\widehat{\text{ATE}}_{\text{High}} - \widehat{\text{ATE}}_{\text{Low}} = -0.054 \quad (95\% \text{ CI: } \pm 0.022)$$

Interpretation

This significant difference supports the presence of treatment effect heterogeneity and confirms that the causal forest captures variation beyond the average effect.

Formal Tests for Heterogeneity by Covariates

To identify which covariates drive heterogeneity, I conduct hypothesis tests splitting clusters by median covariate value and comparing treatment effects.

Method:

- Compute doubly-robust score for each observation.
- Aggregate scores by cluster.
- Split clusters into high vs. low by each covariate.
- Perform two-sample t-test.

Significant Findings:

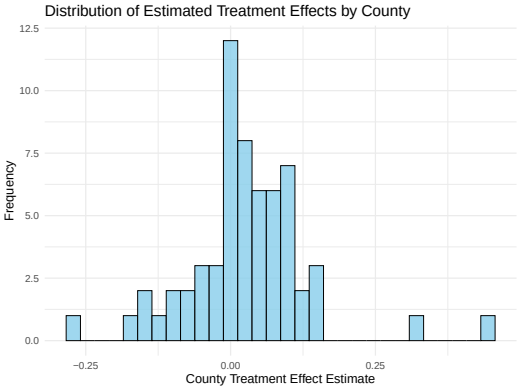
- **Expenditures on schools:** Higher spending $\rightarrow$ lower treatment effects. Estimated difference $= -0.064$ ($p = 0.017$).
- **State money to charities:** Higher support $\rightarrow$ lower treatment effects. Estimated difference $= -0.055$ ($p = 0.040$).

County-wise Heterogeneity of Treatment Effects

Explored the heterogeneity of treatment effects across counties. Each bar shows the average treatment effect within a county cluster.

Interpretation:

- Treatment effects vary substantially across counties.
- Most estimates cluster near zero.
- Some counties show positive or negative effects.
- Indicates spatial heterogeneity in program impact.



Source: data from Aizer et al. (2016); author's own calculations.

Sanity Check: Re-estimating the ATE by County Means

Computed a sanity check by averaging treatment effects across counties and comparing them to causal forest estimates.

Findings:

- County-level mean ATE: approx. 2 years increase in lifespan.
- Causal forest ATE: approx. 1.55 years (95% CI excludes zero).
- Consistent positive effects across methods.

Comparison Table:

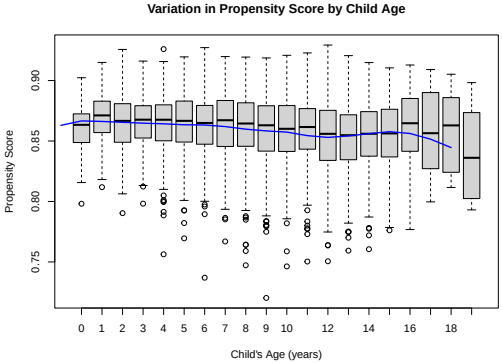
Method	ATE (log)	95% CI (log)	ATE (years)	95% CI (years)
County mean	0.03	[0; 0.054]	2.00	[0.03; 4.03]
Causal forest	0.02	[0.005; 0.037]	1.55	[0.35; 2.76]

Variation in Propensity Scores Across Child Age

Examined how estimated treatment probability varied with child age.

Findings:

- Propensity stable across ages 0–13 (median 0.85–0.88).
- Modest decline for ages 14–18.
- Interquartile ranges overlap—no severe imbalance.



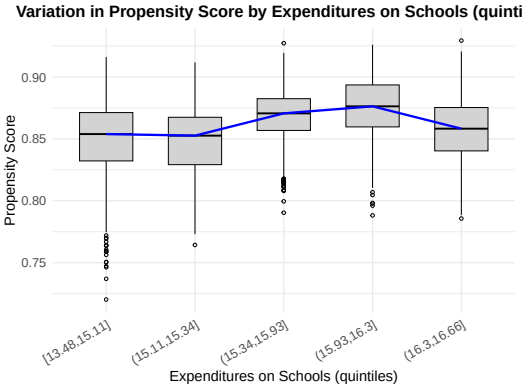
Source: data from Aizer et al. (2016); author's own calculations.

Variation in Propensity Scores Across School Expenditures

Grouped counties by quintiles of school spending to assess correlation with treatment probability.

Findings:

- Slightly higher propensity scores in high-expenditure counties.
- Differences modest; distributions broadly overlap.
- No dominant selection effect.



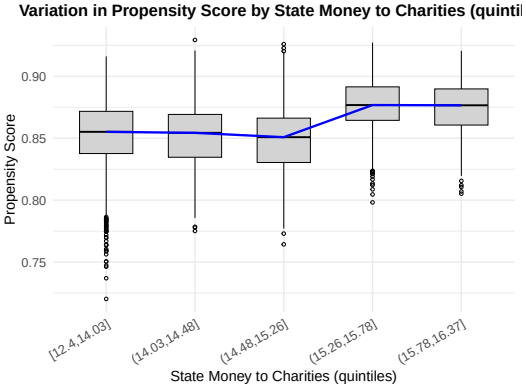
Source: data from Aizer et al. (2016); author's own calculations.

Variation in Propensity Scores Across Charity Spending

Examined treatment probability by quintiles of state spending on charities.

Findings:

- Mild positive association with propensity scores.
- Distributions overlap substantially.
- No strong threshold effect.



Source: data from Aizer et al. (2016); author's own calculations.

Analysis Ignoring Clustering

Estimated the causal forest ignoring clustering within counties to assess how clustering affects precision and detection of heterogeneity.

Findings:

- ATE = 0.022 (95% CI $\pm$0.014), close to clustered estimate.
- Standard error smaller—ignoring clustering likely overstates precision.
- **Calibration test:** *Mean prediction coefficient* = 0.945 (significant) — model captures average effect well. *Differential prediction coefficient* = -1.058 (not significant) — heterogeneity estimates mainly noise.
- **ANOVA:** No significant residual county-level differences ($F = 0.8$, p=0.867).

Interpretation and Intuition

- Ignoring clustering leaves mean estimates unbiased but understates uncertainty.
- The calibration test shows heterogeneity is not reliably detected in this specification.
- ANOVA shows no significant residual county-level differences after covariate adjustment.

Estimated Treatment Effects by County (No Clusters)

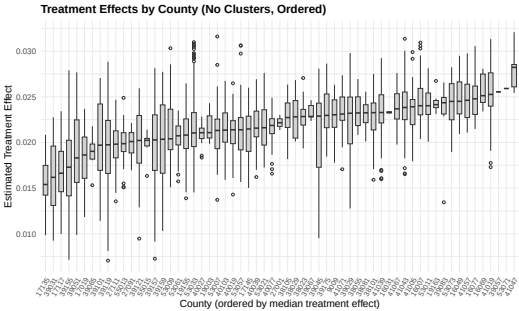
Distribution of estimated treatment effects by county without clustering.

Description:

- Each boxplot shows the distribution of CATEs within a county.
- Counties are ordered by median estimated treatment effect.
- Most medians around 0.02 with modest variation.

Interpretation and Intuition

- Ignoring clustering yields modest variation in treatment effects across counties.
- No county shows substantially higher or lower effects.



Source: data from Aizer et al. (2016); author's own calculations.

Analysis Without Orthogonalization

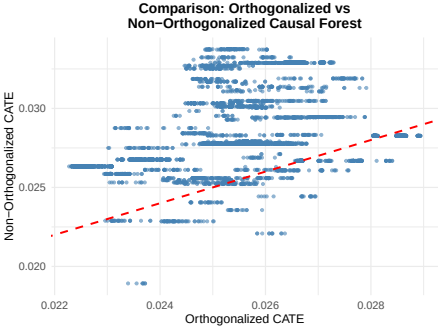
Estimated a causal forest without modeling the propensity score (no orthogonalization).

Findings:

- ATE = 0.027 (95% CI ± 0.027).

Interpretation and Intuition

- Estimates are positively correlated but diverge at higher predicted CATEs.
- Omitting orthogonalization leads to bias and instability in treatment effect estimation.
- Orthogonalization improves robustness and yields more credible ATEs and CATEs.



Comparison of Orthogonalized vs. Non-Orthogonalized CATE Estimates. The dashed red line denotes the 45-degree line. Source: data from Aizer et al. (2016); author's own calculations.

Conclusion

This paper examined the long-run effects of the Mothers' Pension program on male children's longevity using causal forests and rich linked data.

Key Findings:

- Receiving MP support increased lifespan by approximately 1.5–2 years on average.
- Benefits were not uniform: children in states with lower public spending on education and charities experienced larger gains.
- Robustness checks showed the importance of orthogonalization and clustering for reliable inference.

Interpretation and Implications:

- Early targeted social transfers can have sustained positive impacts on life outcomes.
- Modern machine learning methods help uncover nuanced treatment effects in historical data.

Future Directions:

- Explore heterogeneous effects using additional covariates.
- Extend analysis to female children.

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Appendix: Summary Statistics I

Summary Statistics for Estimation Sample (Matched Data)

Variable	Mean Treated	Mean Control	Difference	SE (Diff.)
Child age (years)	8.31	8.83	-0.52	0.14
Length of family name	6.36	6.34	0.02	0.06
Number of kids in family	3.73	3.54	0.19	0.07
Age oldest child	11.47	11.86	-0.39	0.15
Age youngest child	4.86	5.63	-0.77	0.13
Divorced or never married	0.031	0.030	0.001	0.006
Husband absent / prison	0.188	0.171	0.017	0.013
Marital status unknown	0.233	0.268	-0.035	0.015
Birth date missing	0.018	0.007	0.011	0.003
Mean SEI	24.96	24.38	0.58	0.11
SD SEI	20.72	20.82	-0.09	0.08
% urban	0.424	0.359	0.064	0.009
% widowed	0.063	0.061	0.002	0.001
% children single mother	0.055	0.056	-0.001	0.001
% low SEI	0.101	0.110	-0.009	0.001
% women working	0.183	0.182	0.001	0.001

Appendix: Summary Statistics II

Variable	Mean Treated	Mean Control	Difference	SE (Diff.)
% children working	0.071	0.066	0.005	0.002
Farm value	94.93	80.84	14.10	1.99
State manufacturing wage ratio	1.10	1.12	-0.02	0.00
School entry age	7.62	7.57	0.04	0.03
Min child labor age	14.58	14.79	-0.20	0.03
Continuation school required	0.370	0.245	0.125	0.015
Total social expenditures	13.60	13.60	0.00	0.05
Charity expenditures	14.89	14.92	-0.03	0.04
School expenditures	15.61	15.50	0.11	0.02
Law requires work	0.561	0.678	-0.117	0.016
Law requires re-apply	0.484	0.574	-0.090	0.017
Allowance first child	17.95	19.54	-1.59	0.33
Allowance second child	7.96	8.09	-0.13	0.09
Child birth year	1913.59	1912.16	1.43	0.22
Log age at death	4.28	4.27	0.01	0.01

Source: Aizer et al. (2016); author's calculations.

Appendix: Heterogeneity of Treatment Effects I

Heterogeneity Across Covariates

Covariate	Mean High	Mean Low	Difference	95% CI Low	95% CI High	p-value
Expenditures on schools	-0.005	0.059	-0.064	-0.116	-0.012	0.017 *
State money to charities	-0.001	0.055	-0.055	-0.108	-0.003	0.040 *
Total social expenditures	0.007	0.047	-0.040	-0.094	0.014	0.139
Allowance for first child	0.003	0.038	-0.035	-0.082	0.012	0.141
Number of children: 8	0.047	0.009	0.038	-0.016	0.092	0.163
Min. age child labor	0.007	0.037	-0.030	-0.080	0.020	0.229
Age of school entry	0.011	0.043	-0.032	-0.086	0.022	0.241
Husband absent	0.011	0.043	-0.031	-0.086	0.023	0.251
Continuation school required	0.041	0.014	0.027	-0.027	0.082	0.318
State manufacturing earnings ratio	0.017	0.037	-0.019	-0.074	0.035	0.478
Marital status missing	0.037	0.018	0.019	-0.035	0.074	0.481
Length of family name	0.036	0.019	0.018	-0.037	0.072	0.513
(Other rows truncated for brevity.)						

Note: * $p < 0.05$.
Source: Aizer et al. (2016); author's calculations.

Appendix: Determinants of Acceptance (Part 1)

	Accepted	ln(amount)	ln(duration)
Child age (years)	-0.00277	0.00177	-0.0139
	[0.004]	[0.007]	[0.017]
Family name length	-0.000319	0.00240	0.00779
	[0.003]	[0.003]	[0.010]
Family size = 2	0.0349**	0.273***	0.331***
	[0.017]	[0.033]	[0.061]
Family size = 3	0.0511***	0.491***	0.511***
	[0.019]	[0.042]	[0.062]
Family size = 4	0.0624***	0.624***	0.545***
	[0.024]	[0.055]	[0.061]
Family size = 5	0.0570*	0.755***	0.578***
	[0.032]	[0.061]	[0.092]
Family size = 6	0.0626**	0.807***	0.588***
	[0.030]	[0.078]	[0.120]
Family size = 7	0.0654*	0.922***	0.573***
	[0.038]	[0.093]	[0.120]
Family size = 8+	0.0704	0.975***	0.586***
	[0.046]	[0.077]	[0.125]
Oldest kid age	-0.00129	-0.00547*	-0.00665
	[0.002]	[0.003]	[0.006]
Youngest kid age	-0.00258	-0.0133***	-0.0178**
	[0.002]	[0.003]	[0.007]

Robust SEs in brackets. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.
Source: Aizer et al. (2016).

Appendix: Determinants of Acceptance (Part 2)

	Accepted	ln(amount)	ln(duration)
Divorced	-0.0415	0.00865	0.0148
	[0.034]	[0.028]	[0.030]
Husband absent	-0.00448	-0.00651	-0.0815***
	[0.010]	[0.017]	[0.026]
Marital status unknown	-0.116	-0.0333	0.0764
	[0.076]	[0.038]	[0.065]
Birth info missing	0.0399	0.0433	0.356*
	[0.039]	[0.065]	[0.211]
Observations	7,859	6,820	3,677
Mean	0.875	5.508	1.340

Robust SEs in brackets. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.
Source: Aizer et al. (2016).